

# Earnings surprise

Directed factor-mimicking portfolio catalogue

## Abstract

This in-sample catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by directed raw Sharpe, highest first. Direction is the sign that makes comparison-window Sharpe non-negative.

## Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes direction and Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only.

## Construction Configuration

| Setting            | Effective value                                                      |
|--------------------|----------------------------------------------------------------------|
| Factors            | 2                                                                    |
| Universe           | shared security screen; refreshed monthly                            |
| Formation filters  | price $\geq 1.0$ ; 21d dollar volume $\geq$ none                     |
| Market-cap screen  | top 60.0% monthly                                                    |
| Signal preparation | within-month winsorize, normalize, then demean                       |
| Composite          | mean available normalized components; factor-specific coverage floor |
| Construction       | one blanket profile                                                  |
| Rebalance / return | month-end / next calendar month simple return                        |
| Costs              | effective configured costs; default 0 bps                            |
| Minimum sample     | 20 stocks/month; 24 common months; 90% factor coverage               |
| Comparison window  | 2006-05-01 to 2026-05-01; factor- and control-complete               |
| Winsor settings    | 1%-99%                                                               |
| Controls           | MKT, SMB, VAL, MOM, BAB, QMJ                                         |

## Returns PCA

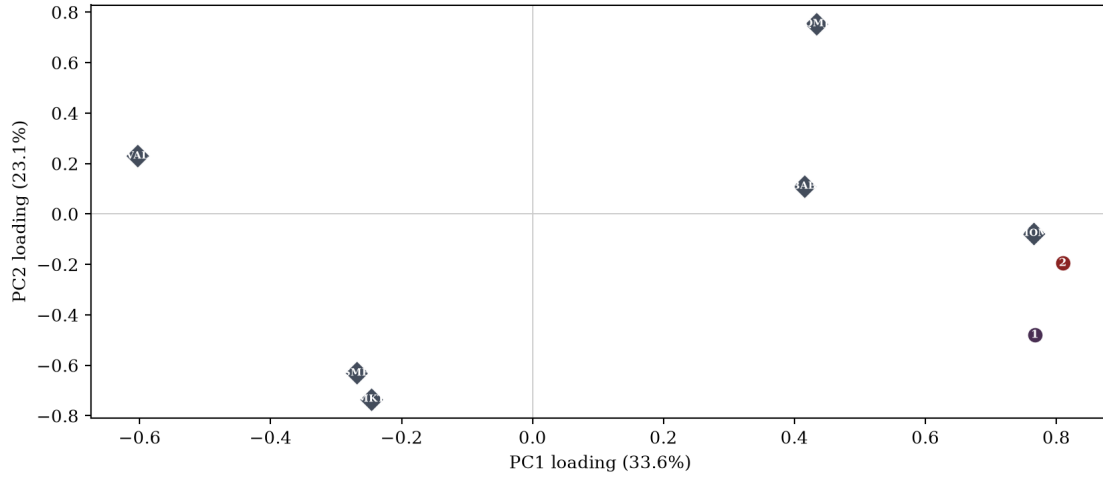


Figure: catalogue factors are numbered by directed-Sharpe index; attribution controls are named;  $N=241$  common months.

# Index

| Idx | Factor             | Dir. | N   |
|-----|--------------------|------|-----|
| 1   | SurprisePercentage | +1   | 241 |
| 2   | Surprise           | +1   | 241 |

# Returns

| Factor             | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|--------------------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| SurprisePercentage | 2.63%     | 4.40%     | 0.60   | 2.43  | -18.50% | 0.299    | 1.23%       | 4.40%      |
| Surprise           | 1.08%     | 4.45%     | 0.24   | 1.04  | -24.60% | 0.288    | -0.96%      | 4.45%      |

# Risk Loadings

| Factor             | MKT            | SMB              | VAL              | MOM            | BAB            | QMJ            | Alpha             | R2    |
|--------------------|----------------|------------------|------------------|----------------|----------------|----------------|-------------------|-------|
| SurprisePercentage | 0.07<br>(3.10) | 0.10<br>(3.10)   | -0.07<br>(-1.76) | 0.11<br>(4.21) | 0.04<br>(1.14) | 0.08<br>(1.97) | 1.06%<br>(1.17)   | 0.302 |
| Surprise           | 0.09<br>(3.23) | -0.00<br>(-0.12) | -0.00<br>(-0.02) | 0.12<br>(4.52) | 0.06<br>(1.19) | 0.16<br>(3.11) | -0.98%<br>(-1.08) | 0.288 |

# Return Contributions

| Factor             | MKT   | SMB    | VAL   | MOM   | BAB   | QMJ   | Alpha  | Total |
|--------------------|-------|--------|-------|-------|-------|-------|--------|-------|
| SurprisePercentage | 0.83% | -0.09% | 0.10% | 0.35% | 0.16% | 0.22% | 1.06%  | 2.63% |
| Surprise           | 1.01% | 0.00%  | 0.00% | 0.37% | 0.20% | 0.47% | -0.98% | 1.08% |

# 1. SurprisePercentage

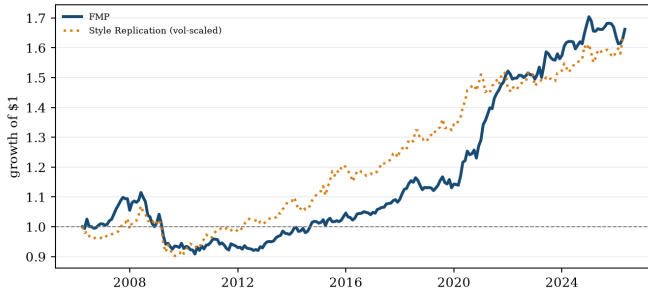


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor             | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|--------------------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| SurprisePercentage | 2.63%     | 4.40%     | 0.60   | 2.43  | -18.50% | 0.299    | 1.23%       | 4.40%      |

| Factor             | MKT            | SMB            | VAL              | MOM            | BAB            | QMJ            | Alpha           | R2    |
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| SurprisePercentage | 0.07<br>(3.10) | 0.10<br>(3.10) | -0.07<br>(-1.76) | 0.11<br>(4.21) | 0.04<br>(1.14) | 0.08<br>(1.97) | 1.06%<br>(1.17) | 0.302 |

| Factor             | MKT   | SMB    | VAL   | MOM   | BAB   | QMJ   | Alpha | Total |
|--------------------|-------|--------|-------|-------|-------|-------|-------|-------|
| SurprisePercentage | 0.83% | -0.09% | 0.10% | 0.35% | 0.16% | 0.22% | 1.06% | 2.63% |

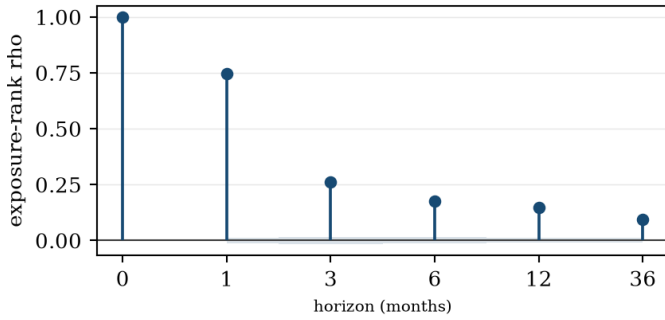


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

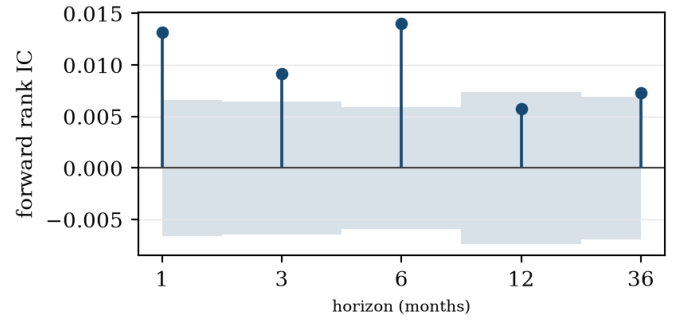


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                            |
|-------------------------------|----------------------------------------------------------------------------------|
| Direction                     | +1                                                                               |
| Common months                 | 241                                                                              |
| Signals                       | surprisePercentage(+1)                                                           |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp |
| Components required           | 1 of 1                                                                           |
| Available / comparison months | 243 / 241                                                                        |
| Available window              | 2006-05-01 to 2026-07-01                                                         |
| Dropped: low breadth / common | 44 / 2                                                                           |
| Stocks, min / median / max    | 28 / 1373 / 2511                                                                 |
| Available-history Sharpe      | 0.62                                                                             |

## 2. Surprise

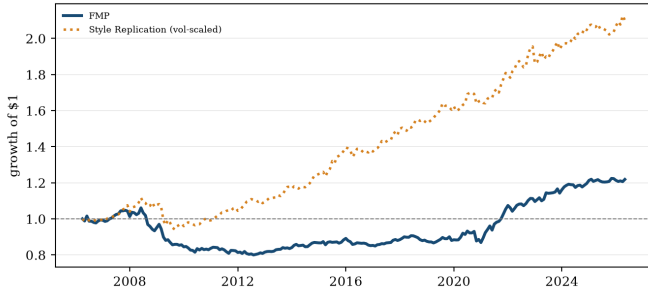


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| Surprise | 1.08%     | 4.45%     | 0.24   | 1.04  | -24.60% | 0.288    | -0.96%      | 4.45%      |

| Factor   | MKT            | SMB              | VAL              | MOM            | BAB            | QMJ            | Alpha             | R2    |
|----------|----------------|------------------|------------------|----------------|----------------|----------------|-------------------|-------|
| Surprise | 0.09<br>(3.23) | -0.00<br>(-0.12) | -0.00<br>(-0.02) | 0.12<br>(4.52) | 0.06<br>(1.19) | 0.16<br>(3.11) | -0.98%<br>(-1.08) | 0.288 |

| Factor   | MKT   | SMB   | VAL   | MOM   | BAB   | QMJ   | Alpha  | Total |
|----------|-------|-------|-------|-------|-------|-------|--------|-------|
| Surprise | 1.01% | 0.00% | 0.00% | 0.37% | 0.20% | 0.47% | -0.98% | 1.08% |

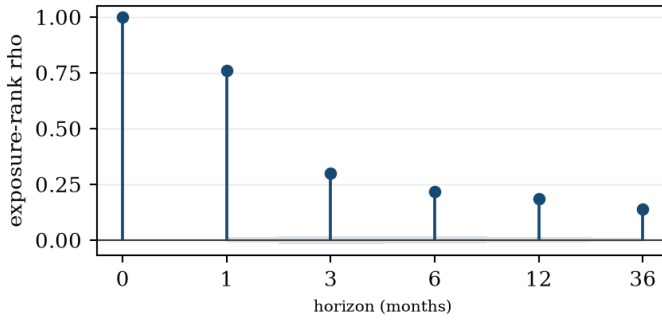


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

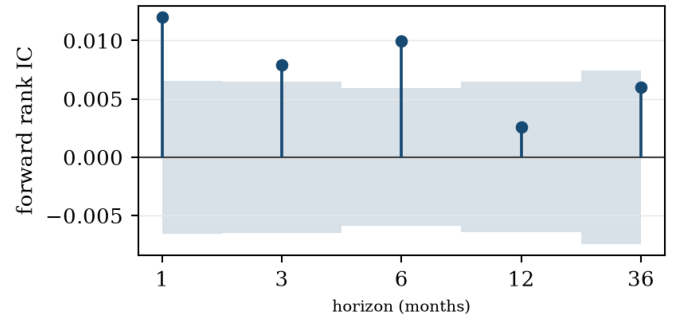


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                            |
|-------------------------------|----------------------------------------------------------------------------------|
| Direction                     | +1                                                                               |
| Common months                 | 241                                                                              |
| Signals                       | surprise(+1)                                                                     |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp |
| Components required           | 1 of 1                                                                           |
| Available / comparison months | 243 / 241                                                                        |
| Available window              | 2006-05-01 to 2026-07-01                                                         |
| Dropped: low breadth / common | 44 / 2                                                                           |
| Stocks, min / median / max    | 28 / 1395 / 2560                                                                 |
| Available-history Sharpe      | 0.29                                                                             |

## Reproducibility

Run ID: 20260829T041531Z\_earnings-surprise\_f4109ffc05.

Gold snapshot: version=20260826T131322Z.

Configuration hash: f4109ffc05.

Symbols requested / loaded: all / 5857.

Seed: 0. Comparison months: 241, 2006-05-01 to 2026-05-01.

Command: C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/  
factor\_model\_runner/run.py--configsrc/experiments/factor\_model\_runner/revenue\_earnings\_surprise.py--n0--seed0.

Direction is selected once from raw comparison-window Sharpe and then applied before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.